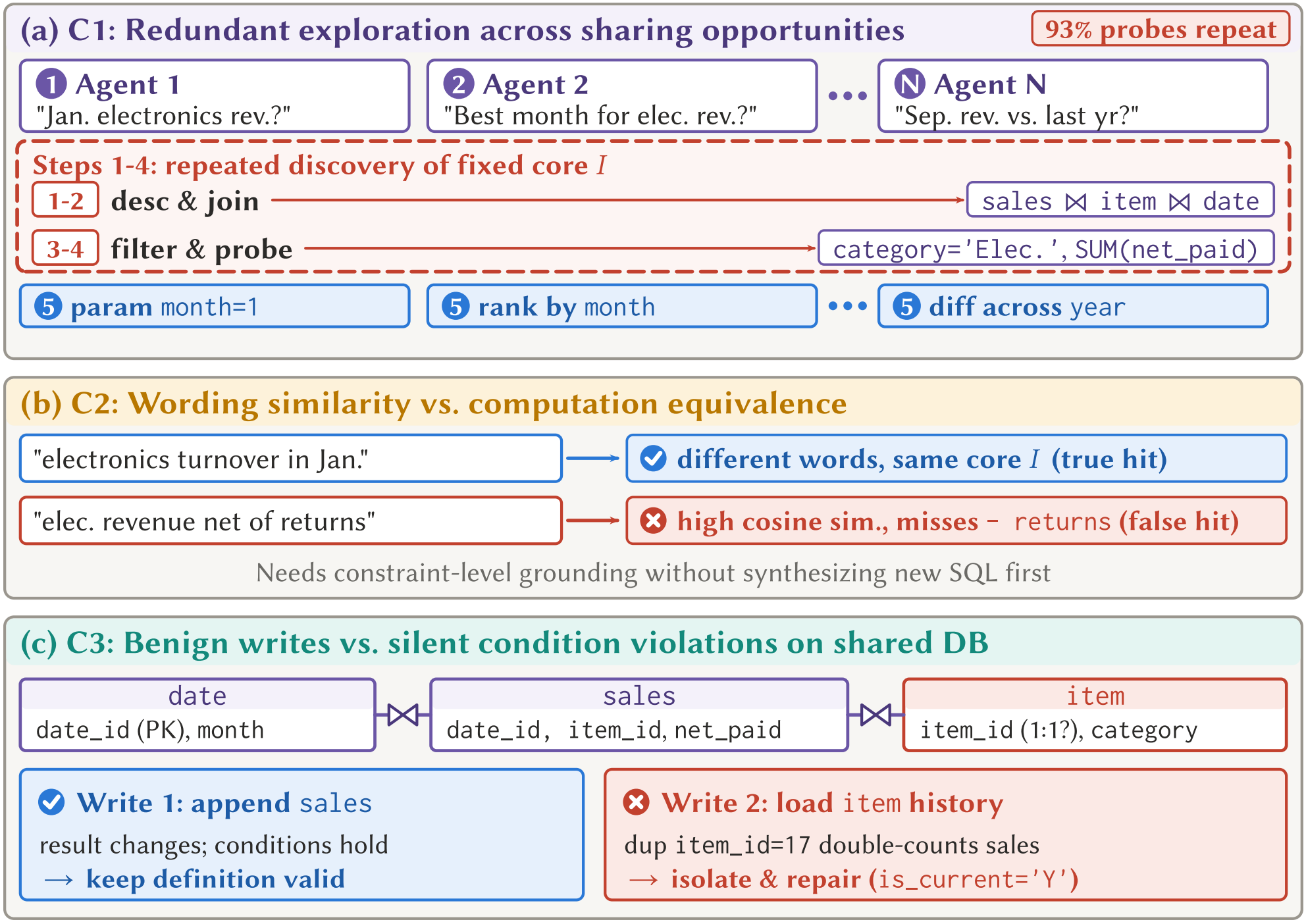

(a) C1: Redundant exploration across sharing opportunities
93% probes repeat
1 Agent 1
"Jan. electronics rev.?"
2 Agent 2
"Best month for elec. rev.?"
N Agent N
"Sep. rev. vs. last yr?"
Steps 1-4: repeated discovery of fixed core I
1-2 desc & join
sales ⋈ item ⋈ date
3-4 filter & probe
category='Elec.', SUM(net_paid)
5 param month=1
5 rank by month
5 diff across year
(b) C2: Wording similarity vs. computation equivalence
"electronics turnover in Jan."
different words, same core I (true hit)
"elec. revenue net of returns"
high cosine sim., misses − returns (false hit)
Needs constraint-level grounding without synthesizing new SQL first
(c) C3: Benign writes vs. silent condition violations on shared DB
date
date_id (PK), month
sales
date_id, item_id, net_paid
item
item_id (1:1?), category
Write 1: append sales
result changes; conditions hold
→ keep definition valid
Write 2: load item history
dup item_id=17 double-counts sales
→ isolate & repair (is_current='Y')